

Treatment of papulopustular rosacea

Inflammatory lesions such as papules and pustules, as well as erythema, occur in subtype II rosacea. In mild manifestations, topical treatments alone can be effective. For more severe cases, systemic treatment or a combination of topical and systemic therapies are recommended.

Topical treatment of papulopustular rosacea

Metronidazole

Metronidazole (MTZ) is a nitroimidazole antibiotic. For subtype II rosacea, 0.75–1% cream or gel is most commonly used and should be applied twice daily. In Europe, multiple metronidazole formulations are available. However, studies have shown no significant difference in efficacy between 0.75% and 1% concentrations, and it remains unclear whether the higher dose offers improved outcomes. The efficacy of MTZ has been confirmed in multiple double-blind studies. Compared to azelaic acid (AZA), MTZ shows similar or slightly inferior results in reducing facial erythema, though it is generally considered better tolerated. In a head-to-head trial with pimecrolimus 1%, both agents reduced erythema equally, but neither improved telangiectasias. The maximum reduction in lesion count is typically observed around 6 weeks. One randomized double-blind trial (n = 51) found MTZ to have similar efficacy to oral (oxy)tetracycline, with improvement seen in 90% of patients in both groups.

Conclusion: Metronidazole is recommended (A) for subtype II rosacea patients.

Tacrolimus

A single study involving 24 patients with subtype I and II rosacea showed no reduction in papule count, although some improvement in erythema was observed.

Conclusion: Tacrolimus is not recommended (D) for subtype II rosacea patients.

Azelaic acid

Azelaic acid is a saturated dicarboxylic acid that inhibits the production of reactive oxygen species (ROS) and suppresses pro-inflammatory cytokines such as IL-1, IL-6, and TNF- α . It also downregulates phosphorylation of ERK1/2 and p38, and inhibits UV-induced NF- κ B activation. Additionally, azelaic acid induces PPAR γ , which helps modulate inflammatory responses. Multiple randomized, double-blind, multicenter studies have demonstrated its efficacy in treating subtype II rosacea, with significant improvements in both erythema and inflammatory lesions.

Conclusion: Azelaic acid is recommended (A) for subtype II rosacea patients.

Ivermectin

Ivermectin is a broad-spectrum antiparasitic agent. In a phase 3, investigator-blinded, randomized, parallel-group trial (n = 962), ivermectin 1% cream demonstrated superiority over metronidazole 0.75% cream in reducing inflammatory lesions. It is now approved and available in Switzerland, many European countries, and the USA.

Conclusion: We recommend the use of topical ivermectin (A) for subtype II rosacea patients.

Pimecrolimus

Pimecrolimus 1%, a calcineurin inhibitor, has been used in the treatment of subtype II rosacea. A placebo-controlled, randomized trial (n = 40) showed

efficacy, though no superiority over metronidazole 1% cream was observed. Pimecrolimus 1% also showed some benefit in case reports of granulomatous rosacea. However, flares and increased facial erythema after alcohol consumption have been reported with calcineurin inhibitor use. There are also reports of pimecrolimus potentially inducing rosacea-like symptoms.

Conclusion: Pimecrolimus can be considered (A) for subtype II rosacea patients, but with caution due to potential adverse effects and risk of flare.